

JTMK Project Report:

Smart Inventory Web System

Project Title:

Smart Inventory Web System for Small Businesses

Team Members:

- Muhammad Hail bin Mohd Ariff(Leader)
- Muhammad Alif Najmi Bin Sahri
- Muhammad Nafiz bin Mohamed Nazarudin
- Muhammad Amirul bin Norizan

Introduction:

The Smart Inventory Web System is a project developed to help small businesses manage their inventory more efficiently. The system allows users to track stock levels, add or remove items, and generate detailed reports in real time. It is designed using full-stack web technologies for a responsive and secure experience.

Objectives:

1. To develop a full-stack inventory management web system.
2. To allow real-time updates of inventory levels.
3. To integrate user login and role-based access control.
4. To generate inventory and sales reports for better decision-making.

Tools and Technologies Used:

- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL
- Development Tools: Visual Studio Code, XAMPP
- Version Control: GitHub

Features:

- User authentication with login system.
- Real-time stock updates and notifications for low inventory.
- Product management with add, edit, delete functionality.
- Report generation in PDF and Excel format.
- Mobile responsive interface.

Implementation Process:

1. Planning: Defined scope and database schema.
2. Design: Created UI mockups and wireframes.
3. Development: Implemented backend and frontend modules.

4. Testing: Performed unit and system tests.
5. Deployment: Launched on local server using XAMPP.

Results:

- System tested successfully with 100+ sample products.
- Response time: ~1.2 seconds per operation.
- Accuracy rate: 98% for data entry and stock updates.

Challenges:

- Ensuring data consistency during concurrent operations.
- Implementing secure login to prevent SQL injection.
- Designing an intuitive UI for non-technical users.

Future Enhancements:

- Add barcode scanning for product entry.
- Integrate cloud storage for remote access.
- Develop a mobile app version for Android and iOS.

Conclusion:

This project provided valuable experience in web development, database management, and user interface design. The system offers a practical solution for inventory tracking and can be further enhanced to support advanced business analytics.

References:

1. W3Schools Web Technologies - <https://www.w3schools.com/>
2. PHP Manual - <https://www.php.net/manual/en/>
3. MySQL Documentation - <https://dev.mysql.com/doc/>
4. Bootstrap Docs - <https://getbootstrap.com/docs/5.0/getting-started/introduction/>
5. GitHub Guides - <https://guides.github.com/>